Group members:

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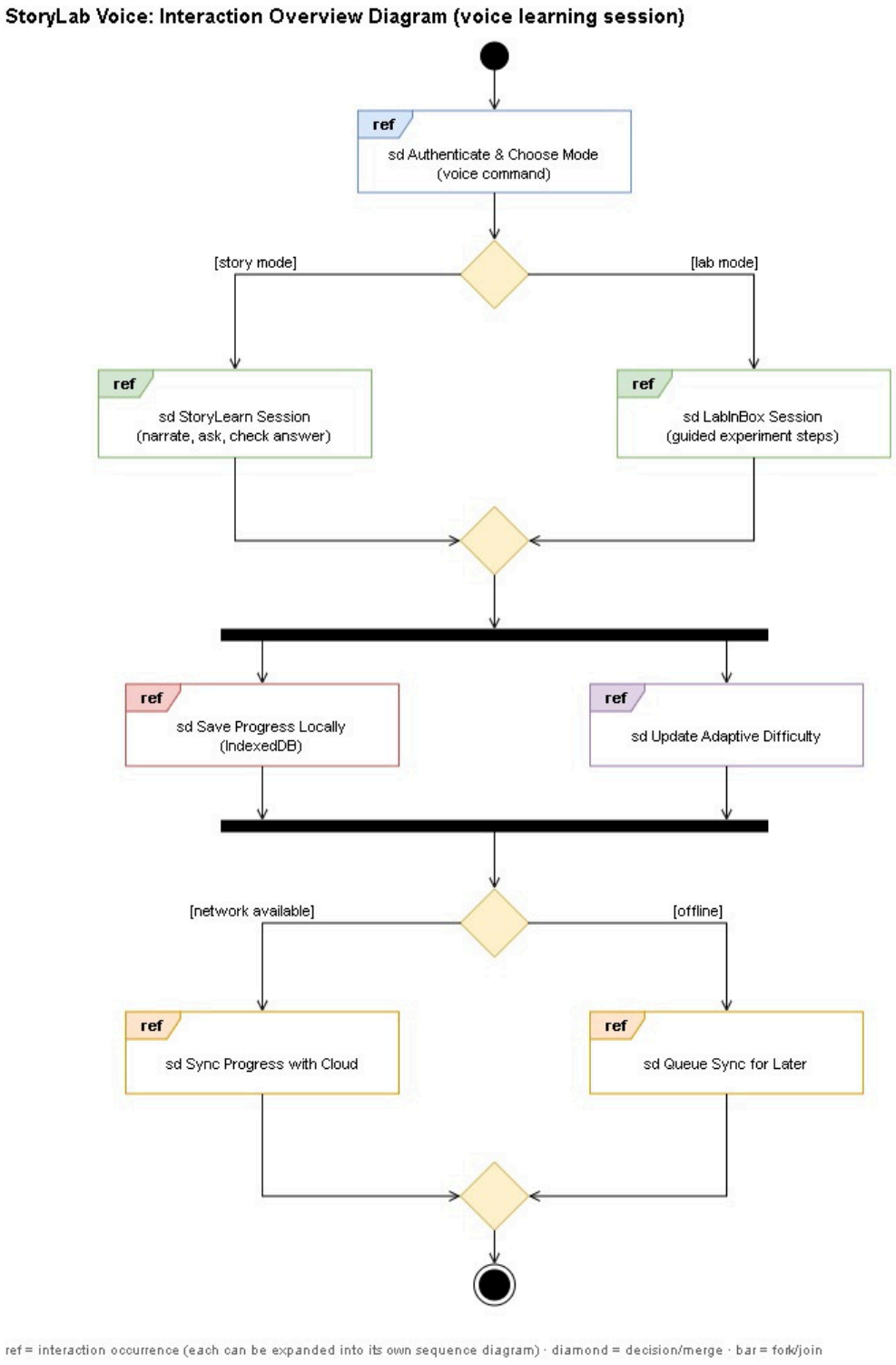
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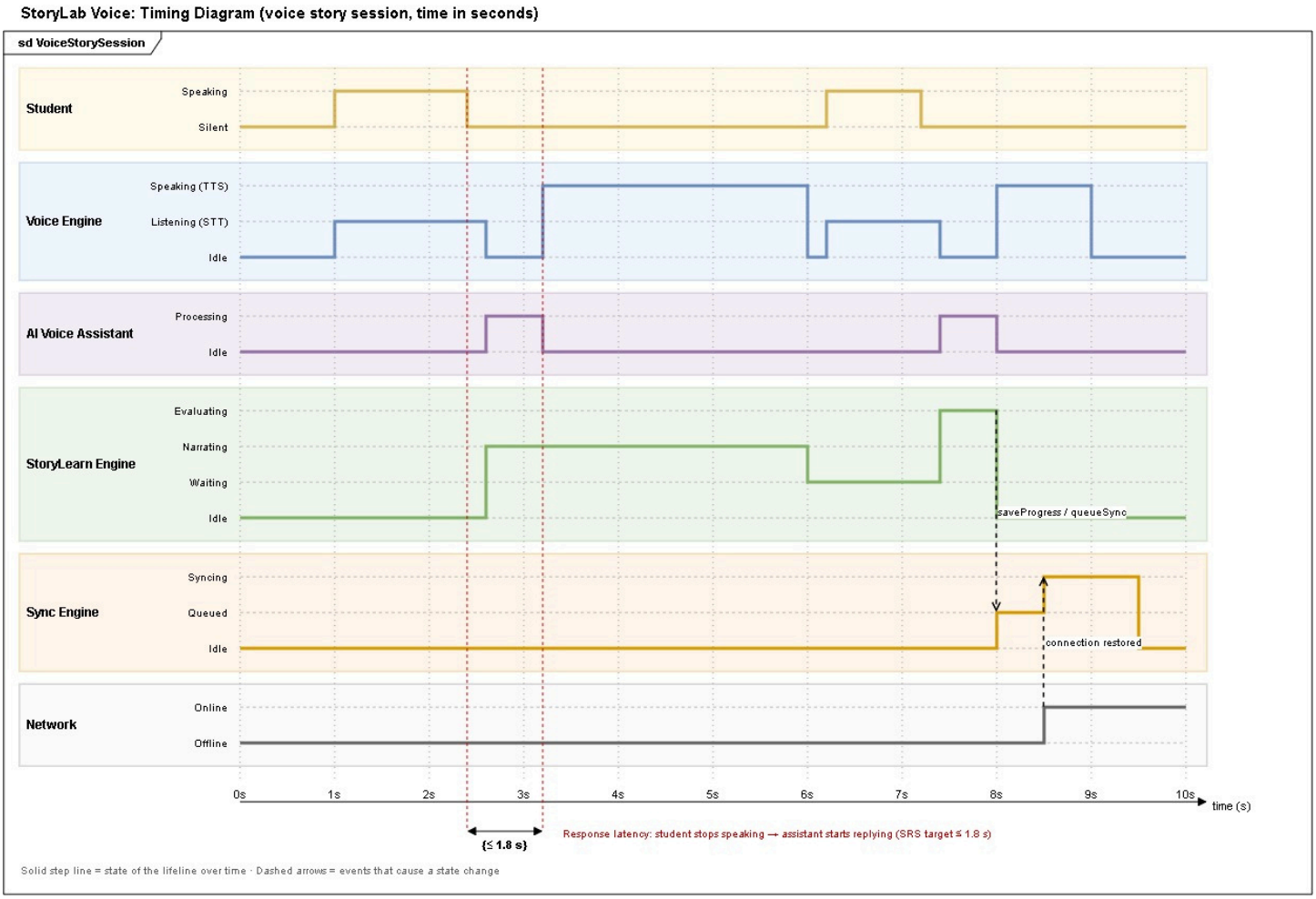
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PRACTICAL NO.07 Designing Complex Interactions with Interaction Overview and Timing Diagram

Interaction Overview diagram:



Timing Diagram:



Interaction overview diagram (storylab_interaction_overview_diagram.drawio)
This diagram gives the high-level flow of a whole learning session. It uses activity-diagram notation, but each step is an interaction occurrence (ref frame) that can be expanded into its own sequence diagram.

The session starts with authentication and mode selection by voice. A decision sends the student to either a StoryLearn session or a LabInBox session, and the two paths merge. A fork then runs two steps in parallel: saving progress locally and updating adaptive difficulty. After the join, a second decision checks the network. If it is available, progress syncs with the cloud. If it is offline, the sync is queued for later. Both paths merge and the session ends.

Timing diagram (storylab_timing_diagram.drawio)
This diagram shows how the state of each part changes over a 10-second voice session. Time runs left to right in seconds. Each lane is one lifeline (Student, Voice Engine, AI Voice Assistant, StoryLearn Engine, Sync Engine and Network), and the step line shows its state at each moment.

For example, the Student alternates between Silent and Speaking. The Voice Engine moves between Idle, Listening (STT) and Speaking (TTS). The assistant is Processing only briefly before each reply.

The {≤ 1.8 s} constraint marks the maximum allowed delay between the student finishing speaking and the assistant starting to reply. The dashed event arrows show causes. Saving progress puts the Sync Engine in the Queued state, and the network coming back online starts the Syncing state. The Network lane stays Offline for most of the session, which shows the offline-first design.